IGBT Test Cell Technical Report

Simulink/Simscape engineering core of a pre-installation IGBT and gate-driver test cell: what the simulations show, what they do not show, and the proposed service procedure.

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0 Verdict

NUMERICAL MODEL VALIDATION

PASS

KCL, energy balance and half-step convergence closed

HARDWARE HEALTH DECISION

INCONCLUSIVE

needs the real Wabtec part numbers, approved limits and measured modules

The project contains two executable Simscape Electrical experiments built on a representative 3.3 kV / 1500 A traction IGBT (Mitsubishi CM1500HC-66XB, not a confirmed Wabtec part):

- an **RL double-pulse test** at 1800 V and about 1.5 kA, recording gate, collector, load, diode and bus quantities;
- a **closed-loop DESAT short-circuit test**, in which the measured V_{CE} passes a blanking, comparator and latch chain that removes the gate command.

The strongest result is an internally consistent electrical experiment that keeps numerical validation separate from hardware acceptance. It does not prove that any real IGBT is healthy.

1 Evidence base

All values in this report were read from the result files below. MATLAB/Simulink produced them; the CSV, JSON and datasheet values were re-checked independently.

FILE	CONTENT
`+testcell/parameters_traction.m`	Representative device and fixture parameters, each tagged datasheet or assumed
`+testcell/run_dynamic.m`, `analyze_dynamic.m`	RL double pulse; KCL, energy, event timing, half-step convergence
`+testcell/run_desat_test.m`	Builds and runs the closed-loop DESAT short-circuit case; exports raw waveforms
`+testcell/run_static_checks.m`, `run_diode_test.m`	$V_{GE(th)}$, I_{CES}, $V_{CE(sat)}$; multimeter-style diode test
`+testcell/build_detailed_models.m`	Generates the detailed presentation models
`results/detailed_simulink_20261017_221616_239/build02/`	Detailed RL re-simulation: `rl_validation.json`, `waveforms/dynamic_rl.csv`, `final/wabtec_igbt_test_cell.slx`. The DESAT model in `final/` was compiled and initialised only, not simulated.
`results/desat_test_traction_20261017_212553_145/`	DESAT run (source of the DESAT numbers)
`results/desat_test_traction_20261018_000313_736/`	Identical DESAT re-run with portable `desat_waveforms.csv` (V_{CE}, I_C, V_{GE}, V_{BUS}, FaultNow, GateCmd, Trip)
`results/static_checks_20261017_193726_230/`, `results/diode_test_traction_20261017_195415_059/`	Static and diode results
`tests/`	Verdict-logic unit tests: 9 of 9 pass (pytest, 18 Oct 2026)
`references/`	CM1500HC-66XB datasheet HVM-1147-B; Wabtec hackathon brief

2 RL double-pulse model

Circuit: ideal 1800 V source → I_{bus} sensor → I_L sensor → load R 0.01 Ω → load L 300 µH → switch node → stray inductance 100 nH → I_C sensor → IGBT → ground. The freewheel diode (with an explicit 11 nF capacitance) connects the switch node back to the bus side of the load. A Simscape Gate Driver (+15 V / −15 V) drives the gate.

ELEMENT	VALUE	SOURCE
Bus voltage / stray L / $R_{G(on)}$ / $R_{G(off)}$	1800 V / 100 nH / 2 Ω / 27 Ω	Datasheet switching test conditions
Gate levels	+15 V / −15 V	Wabtec slides 5 and 8; datasheet
Load R / load L	0.01 Ω / 300 µH	Assumed (sets ≈ 1.5 kA at the second turn-on)
Driver delay / internal R_G	50 ns / 0.5 Ω	Assumed
Diode knee / slope resistance	0.5 V / 1.2 mΩ	Knee assumed inside slide 5 range; keeps datasheet V_{EC} 2.30 V at 1500 A
Solver	ode15s, max step 0.2 µs	Half-step re-run as convergence check

Why 1800 V, 300 µH and 250 µs. The current slope during the first pulse is about V/L:

```
di/dt ≈ 1800 V / 300 µH ≈ 6 A/µs → ΔI ≈ 6 A/µs × 250 µs ≈ 1500 A
```

So the first pulse (50–300 µs) is sized to reach the 1.5 kA rating of the representative device; the simulation reaches 1479 A at 299 µs.

TIME	COMMAND	WHAT HAPPENS (SIMULATED)
0–50 µs	OFF	V_{GE} ≈ −15 V, V_{CE} ≈ 1800 V, I_C ≈ 0
50–300 µs	ON	Inductor current builds: 293 A at 100 µs, 890 A at 200 µs, 1479 A at 299 µs (V_{CE} 2.64 V)
300–400 µs	OFF	Current commutates to the diode: at 350 µs I_L 1577 A, I_C 106 A, I_D 1470 A; at 399 µs I_C 0.79 A, I_D 1573 A
400–500 µs	ON	Hard turn-on against the conducting diode; I_C 2144 A at 499 µs
> 500 µs	OFF	Current returns to the diode

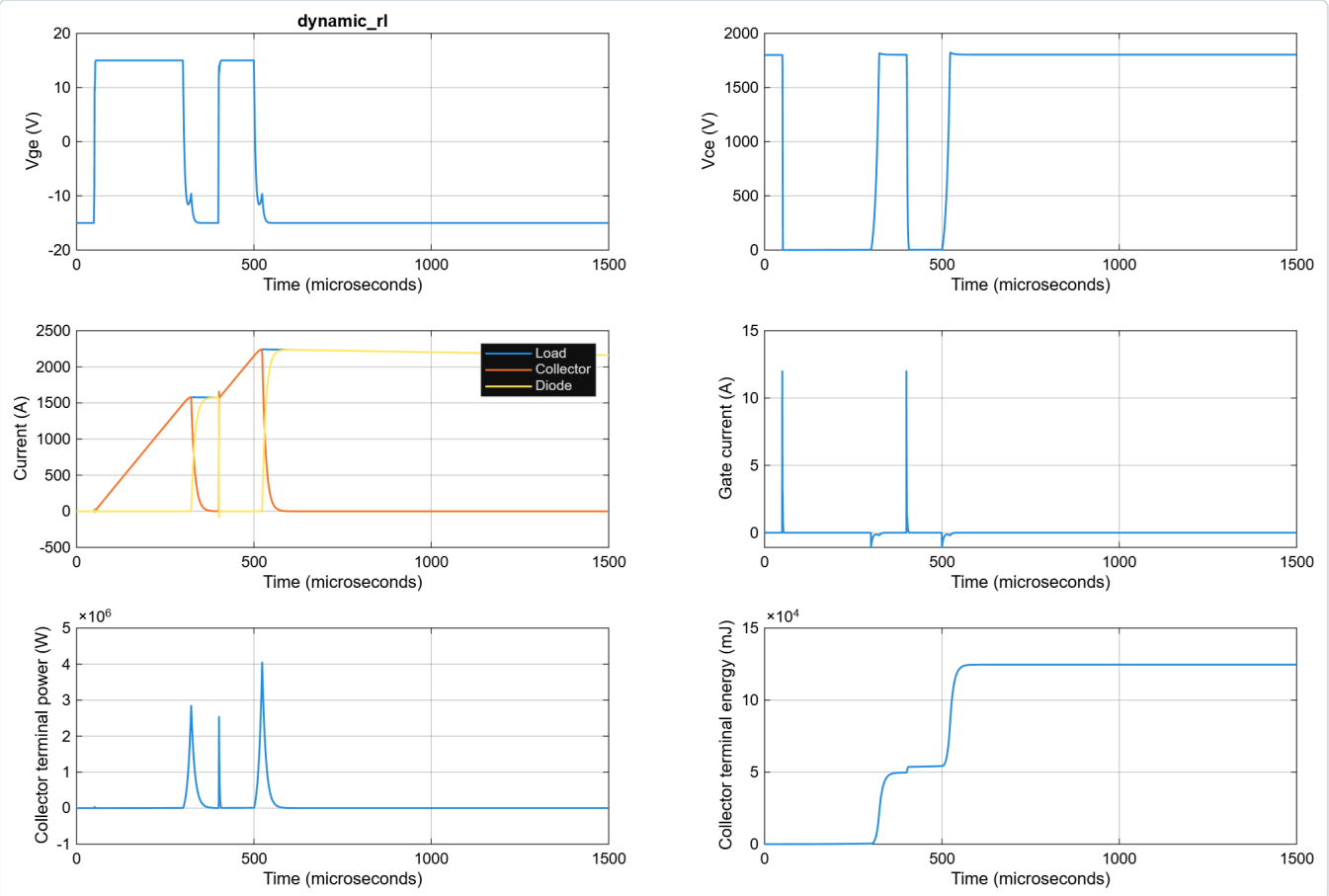


Figure 1. RL double-pulse waveforms, `build02/waveforms/dynamic_rl.png` (SIMULATED).

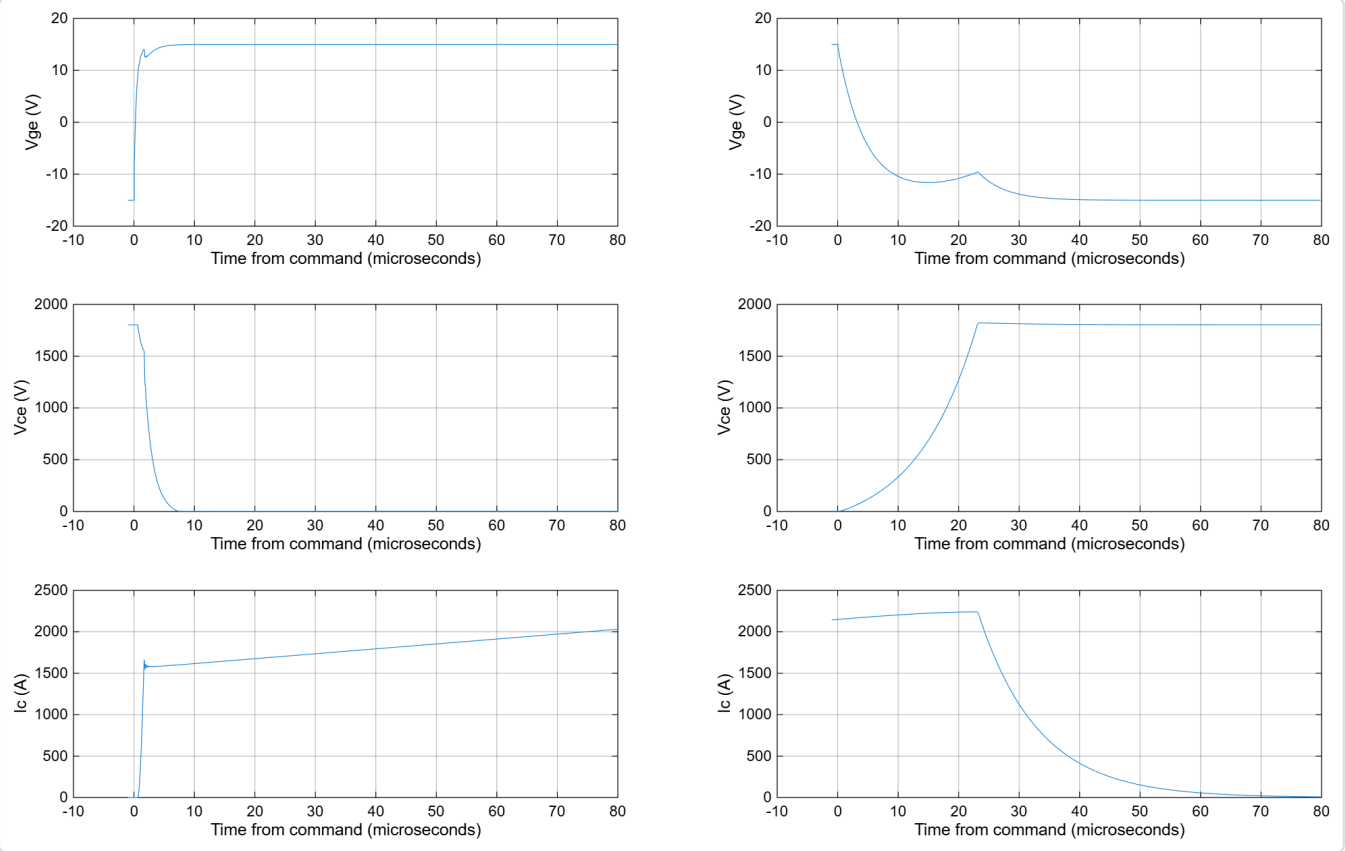


Figure 2. Switching transients of the same run, `build02/waveforms/dynamic_rl_transients.png`.

Numerical consistency

QUANTITY	VALUE
Peak I_C / peak V_{CE} / peak I_G	2240.28 A / 1821.97 V / 12.00 A
Bus energy	888.031 J
Load R / stray R / DUT terminal / diode terminal	56.230 / 0.073 / 124.615 / 6.896 J
Load inductor stored-energy increase	700.218 J
Energy-balance residual	-5.66×10^{-4} J (6.37×10^{-7} relative)
Max KCL residual $\lvert I_L - I_C - I_D \rvert$	1.46×10^{-8} A
Half-step change of event energies	$\leq 1.5 \times 10^{-5}$

This shows the numerical solution and the sensor topology are self-consistent. It does not show that the semiconductor parameters are physically right.

Switching events

EVENT	V_{CE} 10–90 %	COMMAND → V_{CE}	I_C 10–90 %	I_C TO 1 %	GROSS TERMINAL ENERGY
First ON, 50 µs	0.698 µs	0.789 µs	—	—	0.019 J
First OFF, 300 µs	15.29 µs	6.82 µs	21.92 µs	70.01 µs	49.18 J
Second ON, 400 µs	3.28 µs	1.20 µs	0.691 µs	—	4.03 J
Final OFF, 500 µs	15.37 µs	6.75 µs	21.90 µs	69.83 µs	70.29 J

The energies are gross collector-terminal integrals over fixed windows. They include conduction and reactive transfer and are not E_{on}/E_{off} as defined in the datasheet.

Turn-off tail. The modelled current fall (≈ 22 µs, 10–90 %) is far longer than the datasheet maximum t_f 0.70 µs (1800 V, 1500 A, T_j 150 °C; the model runs at 25 °C). The tail parameter is assumed; three calibration attempts stopped the solver, so simulated turn-off timing is not a physical prediction.

Gate charge. Integrating I_G over the second turn-on (400 µs, ≈ 1.6 kA) gives $Q_G \approx 7.44$ µC, about 9 % below the datasheet typical 8.2 µC (1800 V, 1500 A, ±15 V, 25 °C). This is a consistency indication, not a calibration.

3 Static and diode checks

CHECK	SIMULATED	DATASHEET (REPRESENTATIVE)	STATUS
$V_{GE(th)}$, V_{CE} 10 V, I_C 150 mA, 25 °C	6.350 V	5.80 – 6.80 V	INSIDE
$V_{CE(sat)}$ 1500 A, 25 °C	2.600 V	typ 2.60 V	CALIBRATION POINT
$V_{CE(sat)}$ 1500 A, 125 °C	3.200 V	typ 3.20 V	CALIBRATION POINT
$V_{CE(sat)}$ 1500 A, 150 °C	3.348 V	typ 3.30, max 3.70 V	PREDICTION INSIDE MAX
I_{CES} 3300 V, 25 °C	—	max 3.0 mA	RECOVERY CHECK FAILED
Diode forward at 1 mA	0.500 V	slide 5: 0.2 – 0.8 V	PASSES BY CONSTRUCTION
Diode reverse at −3 V	−0.03 µA	slide 5: open	OPEN

The static suite is **PARTIAL**: the model's leakage floor (≈ 3 µA at 3300 V) prevents recovering the 25 °C I_{CES} calibration value, and the tolerance was not loosened. $V_{CE(sat)}$ at 25 and 125 °C reproduces the values the model was calibrated on, so it is not independent proof. The diode knee was set to 0.5 V, so the diode test demonstrates the automated test pipeline rather than a diode property.

4 Closed-loop DESAT short circuit

Type-I short circuit: the gate is switched on into a short across the 1800 V link (only the 100 nH stray inductance limits the current), the case slide 8 describes as a failed companion IGBT. The DC link is first ramped from 10 V to 1800 V over 8 µs, because starting the network in the 1800 V equilibrium stalled the solver.

```
FaultNow = (V_CE > 9 V) AND BlankingDone(7.5 µs) ; Latch = FaultNow OR Latch_p
```

Settings (all assumed): threshold 9 V, blanking 7.5 µs, detector sampling 50 ns, gate command at 20 µs for 30 µs, stop at 70 µs. The 7.5 µs blanking follows from the healthy double pulse, where V_{CE} needed just over 6.99 µs to fall below 9 V; 5 µs blanking false-tripped. It is a simulation choice, not a Wabtec value.

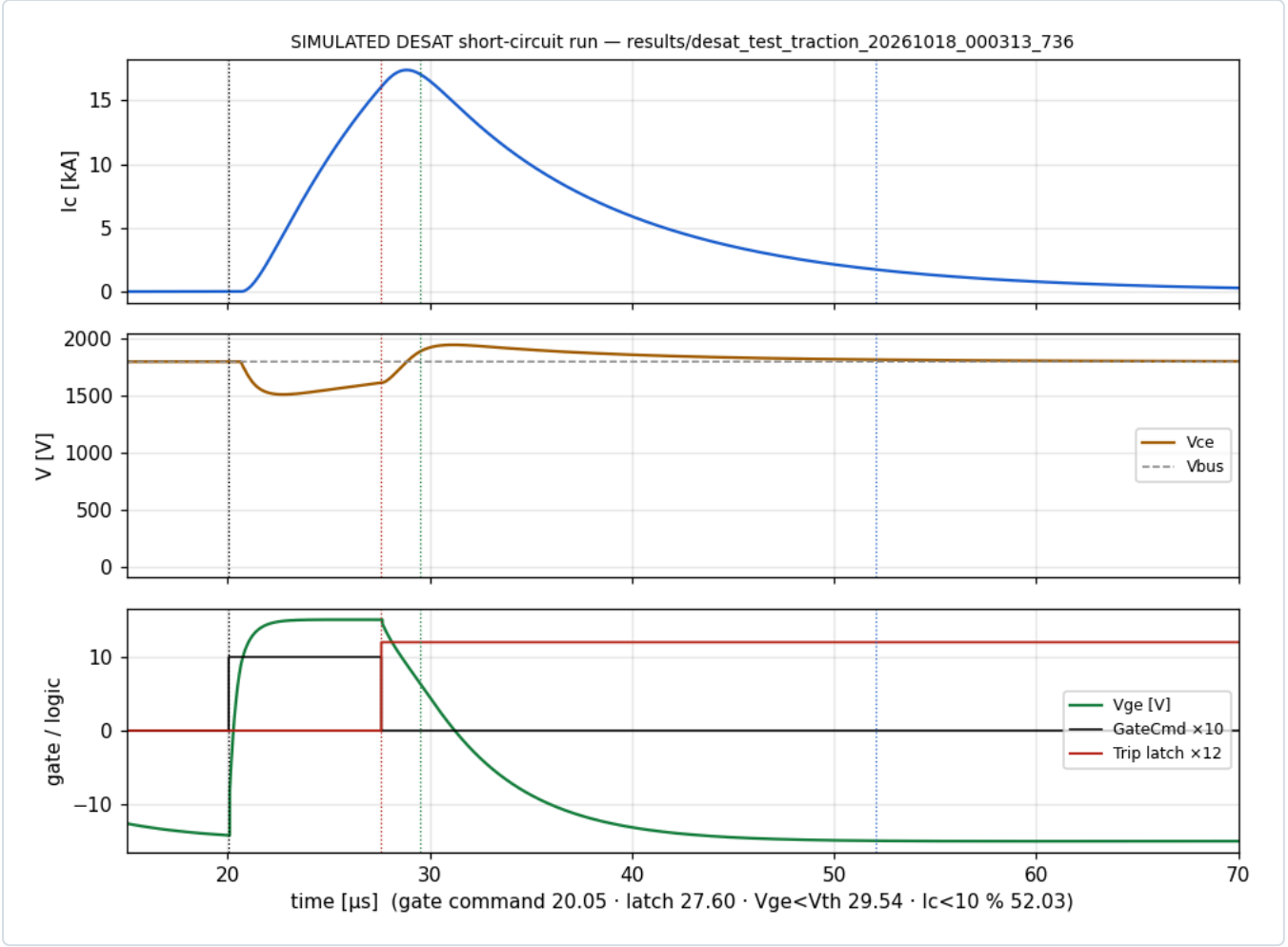


Figure 3. Raw DESAT waveforms from `desat_test_traction_20261018_000313_736/desat_waveforms.csv` (SIMULATED). Dotted lines: gate command, latch, V_{GE} below threshold, I_C below 10 % of peak.

EVENT	TIME	FROM GATE COMMAND	FROM DETECTION
Gate command on (DC link 1800.0 V)	20.05 µs	0	–
FaultNow first asserted	27.55 µs	7.50 µs	–
Latch set, gate command removed	27.60 µs	7.55 µs	0
Peak I_C 17.40 kA	28.85 µs	8.80 µs	1.25 µs
V_{GE} below V_{th} (6.3 V)	29.54 µs	9.49 µs	1.94 µs
I_C below 10 % of peak	52.03 µs	31.98 µs	24.43 µs

Peak V_{CE} after the trip 1947.4 V; short-circuit terminal energy 434.5 J; half-step change 0.04 %. The validation JSON measures from the 20.00 µs command edge (trip 7.60 µs, I_C < 10 % at 32.02 µs); the CSV gate command appears one 50 ns sample later.

The slide 8 notes count the 10 µs from fault detection. The protection logic and driver pull the gate below threshold 1.94 µs after detection, but with the modelled tail the current is below

10 % only 24.4 µs after detection. The run therefore demonstrates protection-loop function, not 10 µs current clearing.

> The 17.4 kA peak is a model value outside the range the model was calibrated on; it is not an expected real short-circuit current and says nothing about device survival. The datasheet t_{pSC} of 10 µs ($V_{CC} \leq 2500$ V, ±15 V, $L_s \leq 100$ nH, $T_j = T_{jop}$) is a withstand rating, not the Wabtec detection-to-turn-off requirement. At gate-on, V_{GE} had settled only to about −14.3 V after the DC-link ramp (Miller coupling); this does not change the result.

5 What to test in service

The service question is whether a removed module and its gate driver are normal enough to reinstall, not a full characterisation of a 3.3 kV device.

TEST	MEASUREMENT	FAULT DETECTED	PRIORITY
C–E integrity	low-voltage resistance / leakage	hard short, open path, severe leakage	Very high
Gate integrity	G–E leakage and continuity	oxide damage, G–E short or open	Very high
Freewheel diode	forward drop, reverse open	diode open or short	Very high
Gate-driver levels	+15 V / −15 V at the gate	weak driver, supply or output fault	Very high
Driver command / status	fiber command, status return	communication fault	Very high
DESAT protection	detection, latch, gate-off timing	protection-chain fault	Very high
Threshold signature	$V_{GE(th)}$	abnormal turn-on	High
Low-energy switching	V_{GE}, V_{CE}, I_C	module cannot switch correctly	High
Temperature sensor	resistance	sensor open or short	Medium
3.3 kV I_{CES}, high-current $V_{CE(sat)}$, E_{on}/E_{off}	engineering-lab measurements	subtle degradation	Optional

Acceptance limits

The representative datasheet values ($V_{GE(th)}$ 5.8–6.8 V, I_{GES} ±0.5 µA at ±20 V, $I_{CES} \leq 3$ mA at 3300 V and 25 °C, $V_{CE(sat)} \leq 3.70$ V at 1500 A and 150 °C) can check the simulation, but they are not Wabtec criteria. Slide values (±15 V gate, diode 0.2–0.8 V and reverse open, ≈ 10 µs

protection, driver ≤ 20 W, busbar leakage ≤ 2 mA) still lack test conditions and tolerances. A missing limit must never become PASS:

- **PASS**: every mandatory result is valid and inside an approved limit.
- **NO-PASS**: a valid measurement violates an approved limit.
- **INCONCLUSIVE / NOT VERIFIED**: part number, test condition, approved limit or valid measurement is missing.

Three-minute service sequence (target)

Benchmark from the brief's speaker notes: about 3.5 minutes per IGBT with PASS / NO-PASS (Wabtec instruction IM-LE-0008, step 14). A 180 s target is credible only with a keyed fixture and automated measurements.

PHASE	PROCEDURE	TARGET
Identification, fixture lock	scan serial, check terminals, lock fixture	15 s
Zero-energy, continuity	confirm discharged; gross C–E / G–E faults	20 s
Diode, temperature sensor	forward / reverse diode, sensor check	20 s
Gate driver standalone	driver supply, ±15 V, command / status, power	30 s
Static IGBT signature	$V_{GE(th)}$, gate leakage, low-voltage C–E leakage	30 s
Low-energy RL switching	pulsed V_{GE} / V_{CE} / I_C functional check	35 s
Protection test	injected or current-limited DESAT fault, latch and gate-off timing	20 s
Discharge, report	open K1, forced discharge, safe-voltage check, result file	10 s
Total		**180 s**

The switching experiment itself lasts 1.5 ms; service time is set by fixture handling, safe-state checks, settling, discharge and reporting. In service, the protection test should use an injected or strongly current-limited fault and record t_{detect}, t_{latch}, $t_{gate\text{-}off}$ and, where meaningful, $t_{current\text{-}clear}$ separately, rather than a full 1800 V hard short on every module.

6 Test-cell concept and cost

Four domains: low-energy power stimulus; gate-driver interface (actual driver, command / status, gate output); isolated measurement of V_{BUS}, V_{CE}, V_{GE}, I_C and status; hard-wired safety (contactor K1, E-stop, door interlock, discharge verification). The laptop sequences the test and writes the report; it is never the only safety layer. K1 closes only when the door is

closed, E-stop healthy, link discharged, fixture locked and no instrument fault is present. The gate driver is tested before any power stage is enabled.

ITEM	EXAMPLE	ROLE	APPROX. COST (USD)
Programmable DC source	B&K 9115B, 80 V / 60 A / 1.2 kW	low-energy switching supply	2,550
DAQ	MCC USB-1608G, 16-bit, 250 kS/s	static and slow channels	459
Precision DMM	B&K 5493C, 6½-digit	static, diode, resistance	1,170
Oscilloscope	Tektronix TBS1102C, 100 MHz, 1 GS/s, 2 ch	gate and protection timing	1,450
Gate-driver / TMC emulator	custom	fiber command / status	500 – 1,000
R-L load, current sensor	custom	controlled switching current	500 – 900
Interlocks, enclosure, K1, discharge	custom	safety chain	800 – 1,500
DUT adapter, busbars	custom	fast, repeatable connection	500 – 1,000
Isolated analog front ends	custom	V_{CE} / V_{GE} / current conditioning	300 – 700
Total, excluding laptop and HV qualification equipment			**≈ 8,230 – 10,730**

Catalogue prices as gathered by the reviewer in October 2026; confirm before presenting. A two-channel scope is a budget anchor: a production cell needs at least four simultaneous isolated fast channels (V_{GE}, V_{CE}, I_C, status). An 80 V stimulus cannot show the slide 5 di/dt range at working voltage; a full 1.8 kV engineering rig (HV source or capacitor bank, precharge and discharge, pulse-rated inductor, HV contactors, rated probes, containment) is quote-based and outside the service-cell budget. Insulation testing needs a separate qualified path; whether a 1000 V insulation tester suffices for the busbar requirement is not known from the brief.

The wiring above is an engineering architecture, not a fabrication drawing. Creepage and clearance, fuse and contactor ratings, protective earth, conductor sizing, containment, EMC, calibration and safety approval (for example under IEC 61010-1) belong to the physical design phase.

7 Limitations and validation plan

LIMITATION	CONSEQUENCE
Representative IGBT, not a confirmed Wabtec part	Datasheet values cannot serve as acceptance limits
Behavioural gate driver, not the Wabtec driver	Fiber, status, driver supply and real protection behaviour untested
Fixed 25 °C, no self-heating	No thermal effects; datasheet switching maxima are given at 150 °C
No diode reverse recovery (datasheet Q_{rr} 880 µC at 25 °C)	Turn-on overshoot and E_{on} underestimated
Turn-off tail not calibrated	Turn-off timing and the DESAT current clearing are not physical predictions
Ideal source and sensors, simplified parasitics	No DC-link sag, bandwidth, noise or fixture ringing
DESAT threshold and blanking assumed; 17.4 kA unvalidated	No statement about real timing or survival
Static suite PARTIAL; diode passes by construction	Not independent validation
No measured Wabtec modules	No known-good / known-bad evidence, no uncertainty analysis

1. Obtain the exact phase-module and gate-driver part numbers, pinout, gate conditions, insulation requirement, protection timing definition and service acceptance criteria from Wabtec.
2. Measure known-good and known-failed modules: V_F, $V_{GE(th)}$, I_{GE}, C–E leakage, V_{GE} on / off, driver and DESAT timing, plus a normalised low-energy $V_{CE}(t)$, $I_C(t)$, $V_{GE}(t)$ signature.
3. Set limits as OEM specification plus measurement uncertainty, from the measured distributions.
4. Compare each bench result with a qualified Wabtec procedure or the known locomotive outcome; report false-pass and false-fail rates.
5. Time the real three-minute procedure with technicians; the phase module weighs about 25 kg.

8 Recommended claim

> We developed and numerically validated the electrical core of a pre-installation IGBT test cell. The simulation demonstrates inductive switching, current commutation, measurement acquisition and closed-loop DESAT protection. On that basis we propose an automated service cell that checks the IGBT and its gate driver in about three minutes and reports

> PASS, NO-PASS or INCONCLUSIVE. Acceptance limits will be set from actual Wabtec modules and OEM data.

SHOWN BY THE SIMULATION	NOT SHOWN
ON / OFF control, current build-up, freewheel commutation, double pulse	That a Wabtec IGBT or gate driver is healthy
KCL and energy closure, solver-step convergence	That protection clears current within 10 µs
DESAT detection, latch and gate inhibit (gate below threshold 1.94 µs after detection)	Survival of the simulated hard short
Automated verdict framework	Real E_{on} / E_{off}; real waveforms; a validated three-minute cycle

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Revision B, 18 Oct 2026. Changes from revision A: corrected file paths (detailed-run files are under `results/detailed_simulink_20261017_221616_239/build02/`); DESAT numbers attributed to `run_desat_test.m` (the detailed DESAT model was compiled only); gate charge attributed to the second turn-on; raw DESAT waveforms added (Figure 3, replaces the timestamp reconstruction); slide 8 "after detection" timing added; datasheet temperature for switching maxima stated; test count updated to 9 of 9; broken download links and copy artefacts removed.

Sources: Wabtec hackathon brief (slides 5, 8, 10, 11 and speaker notes); Mitsubishi CM1500HC-66XB datasheet HVM-1147-B (Dec 2025); project result files listed in section 1.